SMART WATER LOGGING, FLOOD PREVENTION AND POST-FLOOD EVENT MANAGEMENT SYSTEM

INDIVIDUAL THESIS

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ABSTRACT

Floods are the leading natural hazard in Sri Lanka, where their health impact lingers post-flood. An estimated 2.2 million people were affected by the 2025 Cyclone Ditwah floods across all 25 districts, and the affected cohort experiences increased risk of leptospirosis, diarrhoeal disease, respiratory infection, and skin disease. However, the national health post-flood screening system relies on slow manual reporting and in-person examination of already overstretched health teams. Existing disaster platforms prioritise warning, rescue, and logistics, but disaster-level disease surveillance tools are not designed for case triage within a post-flood response flow. This thesis outlines the Post-Flood Disease Risk Management Module of group project R26-IT-031. Two models were trained and integrated into the shared web platform, following a design science research paradigm. The first is a scikit-learn pipeline which classifies post-flood health risk as Low, Medium, or High based on reported symptoms, flood exposures, demographic, age, and pre-existing risk factors. The second is an image-classification service with eight configured output labels (Cellulitis, Dengue, Dermatitis, Eczema, Impetigo, Leptospirosis, Ringworm, and Scabies); the class configuration alone does not establish model architecture, training labels, or clinical diagnostic validity. The evaluated eight-label image feature extraction and SVM classifier achieved 76.12% accuracy and 0.83 macro F1 on 959 test images. Symptom-risk accuracy, High-risk recall, inference latency, and clinical validity remain unverified. Stakeholder feedback was not provided; no stakeholder acceptance claim is made. This work contributes a deployable pair of inference services with explicit constraints, schema safeguarding and a defined integration contract, demonstrating that affordable machine learning can help target post-flood medical treatment. The module provides preliminary decision support and must not be treated as a clinical diagnostic tool.

Keywords: post-flood health, disease risk prediction, skin rash classification, convolutional neural network, disaster management

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LIST OF ABBREVIATIONS

Abbreviation	Meaning
AI	Artificial Intelligence
API	Application Programming Interface
CNN	Convolutional Neural Network
CORS	Cross-Origin Resource Sharing
DMC	Disaster Management Centre
DSR	Design Science Research
EWARN	Early Warning Alert and Response Network
FN	False Negative
FP	False Positive
FR	Functional Requirement
HTTP	Hypertext Transfer Protocol
JSON	JavaScript Object Notation
ML	Machine Learning
NFR	Non-Functional Requirement
NGO	Non-Governmental Organisation
OOD	Out-of-Distribution
PDPA	Personal Data Protection Act
RBAC	Role-Based Access Control
REST	Representational State Transfer
RGB	Red, Green, Blue
SaaS	Software as a Service
SDG	Sustainable Development Goal
SUS	System Usability Scale
SVM	Support Vector Machine

Abbreviation	Meaning
TN	True Negative
TP	True Positive
WHO	World Health Organization

1 INTRODUCTION

1.1 Overall project context

Floods are by far the most common, and the most devastating, natural disaster in tropical developing countries. In Sri Lanka, floods are primarily caused by monsoon rains, watercourses overrunning their banks and an inadequate urban drainage system, and they occur with depressing regularity in districts such as Ratnapura, Kalutara, Gampaha, Colombo, Galle and Matara. In November 2025, cyclone Ditwah led to a level of flooding and landslides the country had not seen in years. The UN situation report, published in January 2026, reports that some 2.2 million people from all 25 districts were affected, 646 people had died, 173 were missing, a maximum of 233,000 people were being housed in 990 protective centres, almost 6,000 houses had been destroyed and over 108,000 houses sustained partial damage [1].

The health impact of such floods extends well beyond the immediate rescue. Evidence and systematic reviews have demonstrated increased risk of faecal-oral disease, vector-borne disease, rodent-borne disease such as leptospirosis, respiratory infection, skin and soft-tissue infection and mental-health problems [2], [3]. Outbreaks tend to occur where displaced populations congregate in crowded shelters with poor water and sanitation [4]. The 2011 flood-related leptospirosis outbreak in Sri Lanka demonstrated how rapidly cases may increase following floods [5] and skin complaints including immersion and irritant dermatitis, fungal infections are among the most common problems reported by responders [6].

Group project R26-IT-031, the Smart Water Logging, Flood Prevention and Post-Flood Event Management System offers one integrated web platform to support the entire flood cycle. As can be seen in Figure 1.1, the web platform has four functional modules: (1) flood and water-logging monitoring and early warning, (2) water rescue operation management, (3) relief and resource management, and (4) post-flood disease risk management. All four modules use the same React web client, and a Node.js application programming interface (API) gateway, MySQL database and role-based access control so they can exchange and use each other's data.

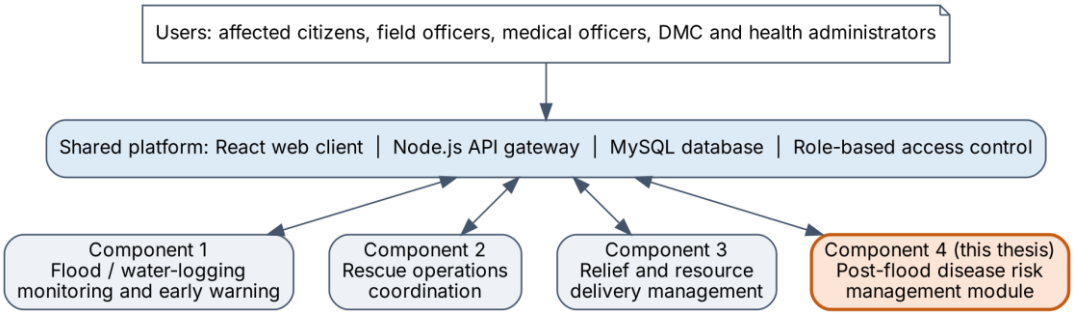


Figure 1.1: Components of the overall R26-IT-031 solution and the position of the individual component

The project presentation lists 1) Flood Map Creation; 2) Drain Water Level Prediction, Real-time Drain Monitoring and Alert System; 3) Intelligent Post-flood Disaster Response and Recovery Management System as the three Components. Please update Figure 1.1 to include these titles.

1.2 Individual component context

This thesis discusses Component 4 - the Post-flood Disease Risk Management Module. This aims to enable medical officers, public health inspectors and disaster-management officers to quickly detect flood-affected people most in need of medical attention, and provide an early overview of trends in emerging health problems in the area.

The initial project design included a symptom-reporting module and a visual analysis component focused on detecting visible effects such as swelling, discoloration, eye redness and fatigue from facial scans. During design, we shifted the visual component from facial scans to images of skin rashes for three reasons: (1) skin problems are one of the most prevalent and visible indicators of post-flooding illnesses [6]; (2) there are existing annotated clinical image datasets of skin conditions but not publicly available, annotated facial data of post-flood illness indicators; and (3) a photo of a specific affected skin region is less intrusive to anonymized facial photos, thus alleviating privacy concerns under Sri Lanka's data legislation [7] (see Section 4.3).

The final component therefore has two intelligent services:

- Symptom-based risk stratification. A scikit-learn pipeline predicts whether the person is at Low/Medium/High risk based on age, gender, duration of symptoms, symptoms reported, flood exposures and risk factors. The Node.js backend calls this python script using a custom Node wrapper in predict.py.
- Image-based classification. The supplied class_names.json configures eight labels: Cellulitis, Dengue, Dermatitis, Eczema, Impetigo, Leptospirosis, Ringworm, and Scabies. The FastAPI inference service (app.py) is described as accepting images and returning a predicted label. The class list is configuration evidence only: confirmation that the saved model has eight aligned output neurons requires inspection of the Keras model and an inference

test. In particular, dengue and leptospirosis cannot be diagnosed from a skin photograph alone.

The outputs of the two services are inputs to the alert, recommendation and dashboard functions of the common platform (see Chapters 4 and 5).

1.3 Problem statement

Following a large flood, health teams need to rapidly identify the displaced population with inadequate staff, laboratory and other diagnostics, and interrupted healthcare services. They largely rely on old-fashioned, manual data collection, arriving at information from conversations and physical examinations. This is a slow process, incompatible between assessors, and provides no roll-up snapshot of where risks are emerging. Therefore some at-risk individuals, like fever post-floodwaters, which may be leptospirosis, and clusters of skin infections, which may be scabies in the camp, are detected too late.

Thus, the problem is: no cheap, fast and integrated means exists that takes information that can be shortly after a flood – self-reported symptoms, exposures, risk factors and photo of a skin lesion–and converts it into a prioritized risk assessment that can be acted on by medical responders within the disaster-management process.

1.4 Research gap

Disaster-management platforms such as Sahana [8] and crowd-sourced crisis-mapping tools such as Ushahidi [9] coordinate people, shelters and resources but do not estimate health risk of individuals. Disease-surveillance systems such as Health Map [10] or WHO's Early Warning Alert and Response approach [11] detect populations outbreaks from case reports but do not triage individuals and are not connected to flood-response operations. Deep-learning dermatology models are highly accurate [12], [13], but trained mostly on dermoscopic or clinical images of cancers and pigmented lesions [14], under-represent darker skin tones [15], [16], and are not optimized for post-flood infectious and inflammatory rashes.

The niche gap is thus the lack of an integrated, individual, post-flood health-risk triage function that (i) combines flood-specific exposure data and symptoms, (ii) incorporates image-based evaluation of typical post-flood skin-injury, and (iii) is implemented as interoperable services within a disaster-management platform and not as a stand-alone clinical instrument.

1.5 Aim and objectives

Aim: To develop and evaluate an intelligent post-flood disease risk management module that assesses individual health risks based on reported symptoms, flood exposure, and relevant risk factors. The module also aims to identify common flood-related skin conditions from photographs and provide the assessment results to medical responders through integration with the R26-IT-031 platform.

Objectives

- Q1: Identify post-flood health conditions, relevant symptoms, flood exposures, and risk factors for health-risk assessment in Sri Lanka through a review of existing literature and consultation with domain experts.

- Q2: Develop and evaluate a machine-learning model to classify individual post-flood health risk into three levels: Low, Medium, and High.

- Q3: Develop and evaluate an image-classification model against the eight configured labels (cellulitis, dengue, dermatitis, eczema, impetigo, leptospirosis, ringworm, scabies), while treating infectious-disease labels as unverified visual proxies rather than confirmed diagnoses.

- Q4: Implement both models as inference services with clearly defined API interfaces, appropriate validation, and error-handling mechanisms.

- Q5: Integrate the inference services with the shared platform to support alert generation, provide relevant recommendations, and display health-risk information on the medical-team dashboard.

- Q6: Evaluate the module in terms of predictive performance, response time, robustness, and usefulness to its intended stakeholders.

1.6 Research questions or hypotheses

This research is guided by the following questions:

- RQ1: How effectively can a machine-learning model use demographic information, symptoms, symptom duration, flood exposure, and relevant risk factors to classify post-flood health risks as Low, Medium, or High, particularly in identifying high-risk individuals?

- RQ2: How reliably can the image-classification system distinguish the eight labels configured in class_names.json under held-out image testing, and what limitations arise for clinically non-specific labels such as dengue and leptospirosis?

- RQ3: Can both models be deployed as inference services that provide sufficiently fast responses and reliable performance for near-real-time use in a flood-response web platform?

- RQ4: How useful and easy to use do medical and disaster-response stakeholders consider the module's outputs when prioritizing post-flood healthcare needs?

For RQ1 and RQ2, the working hypothesis is that both models will perform better than a majority-class baseline when evaluated using appropriate performance metrics. In particular, the health-risk classification model will be assessed for its ability to identify high-risk individuals, while the skin-rash classification model will be evaluated on its ability to distinguish between the three target categories.

The hypothesis will be assessed using the evaluation methods and performance measures described in Section 3.5.

1.7 Scope, Limitations and Assumptions

Scope

This research focuses on developing a post-flood health-risk management module that collects information about symptoms, flood exposure, and relevant risk factors to classify individual health risks into Low, Medium, and High categories. It also includes a skin-rash image classification model with three target classes, along with inference services and integration with the shared platform's alert and dashboard features.

The scope of this research is limited to the health-management component of the overall Flood Guard 360 system. Hardware-based drain monitoring, flood forecasting, rescue logistics, and relief distribution are handled by other components of the group project and are therefore outside the scope of this thesis.

Limitations

Several limitations should be considered when interpreting the results of this research.

- Decision-support purpose: The module is intended to support preliminary health-risk assessment and triage rather than provide a medical diagnosis. It cannot confirm conditions such as leptospirosis, dengue, or other diseases. All outputs should be reviewed by a qualified healthcare professional before any clinical decisions are made.

- Limited image classification: The uploaded class_names.json lists eight output categories: Cellulitis, Dengue, Dermatitis, Eczema, Impetigo, Leptospirosis, Ringworm, and Scabies. The model architecture and index alignment have not been independently verified from the saved .keras artefact. In particular, neither dengue nor leptospirosis is confirmable using a skin photograph alone; outputs must not be presented as definitive disease diagnoses. Images outside the configured set may also be misclassified.

- Dataset limitations: The available inference scripts do not provide sufficient evidence to establish the origin or clinical validation status of the datasets used for training. Although a dataset containing 1,500 records was previously discussed as a possible source for symptom-based risk classification, it should not be identified as the actual training dataset unless this can be confirmed through the training notebook or the dataset itself. Furthermore, the available data may not adequately represent the diversity of Sri Lankan skin tones, lighting conditions, or populations affected by floods.

- Limited field evaluation: Testing during an actual flood event was not possible within the project period. Therefore, the evaluation was limited to held-out data and stakeholder walkthroughs, where applicable. The results may not fully reflect the module's performance under real-world emergency conditions.

Assumptions

The development and intended operation of the module are based on several assumptions. Users or field officers are expected to provide symptom and exposure information that is reasonably accurate. Access to a smartphone camera and intermittent internet connectivity is also assumed at shelters or field locations where the module may be used. In addition, the shared platform is expected to provide appropriate authentication mechanisms and secure data storage to protect sensitive health information.

1.8 Individual Contribution

As part of the group project, the author was responsible for the development and integration of the post-flood health-risk management module. The individual contributions included the following activities:

- Requirements analysis: Identified the requirements of the health module, including the symptoms, flood exposures, and risk factors used for assessment, as well as the definitions of the Low, Medium, and High risk categories.

- Model development and evaluation: Prepared the data, trained and tuned the symptom-based health-risk classification model (flood_risk_model.pkl) and the skin-rash image classification model (final_post_flood_rash_model.keras), and evaluated their performance.

- Backend service implementation: Developed the FastAPI microservice for skin-rash classification (app.py) and the risk-prediction bridge script (predict.py). This work included input validation, schema checks, model-version checks, and error-handling mechanisms.

- System integration: Defined the integration requirements for communication between the health module, the group's Node.js backend, and the React dashboard. The implementation of individual React dashboard screens should be verified against the project repository before specific frontend contributions are attributed to the author.

- Testing and validation: Conducted testing and performance measurements for the health module and contributed to stakeholder validation where applicable.

Shared components, including the common database, authentication mechanisms, and overall platform deployment, were developed as part of the wider group project. These contributions are acknowledged as collaborative work and are not presented as the author's individual achievements.

1.9 Thesis structure

This thesis is organized into nine chapters. Chapter 2 reviews the existing literature on flood-related health risks, machine-learning-based health triage, and image-based skin disease classification. It also identifies the research gap addressed by this study. Chapter 3 explains the research methodology, including the datasets, evaluation metrics, and ethical considerations. Chapter 4 presents the system requirements, overall architecture, and key design decisions. Chapter 5 describes the implementation of the proposed health-risk management module, its integration with the wider platform, and the testing process. Chapter 6 presents the results obtained from the evaluation, while Chapter 7 discusses these findings in relation to existing research and examines the study's limitations. Chapter 8 explores the potential for commercialization and the long-term sustainability of the proposed solution. Finally, Chapter 9 summarizes the research, draws conclusions, and recommends directions for future work.

2 LITERATURE REVIEW AND RELATED WORK

2.1 Review method and scope

The literature review was conducted using a structured narrative approach. Relevant studies were identified through searches of IEEE Xplore, PubMed, ScienceDirect, the ACM Digital Library, and Google Scholar. Search terms included combinations of keywords such as “flood,” “post-disaster,” “health,” “leptospirosis,” “skin disease,” “dermatology,” “deep learning,” “CNN,” “triage,” “machine learning,” and “disaster management system.” Grey literature published by the World Health Organization (WHO), other United Nations agencies, and the Sri Lanka Disaster Management Centre was also consulted to provide additional context on disaster-related health risks and existing response mechanisms.

Sources were selected based on their relevance to three main areas: epidemiological evidence on flood-related diseases, machine-learning approaches to health-risk triage and image-based skin disease classification, and the development or deployment of disaster-management and disease-surveillance systems. Priority was given to peer-reviewed publications and studies that provided clearly documented evaluation methods and results. The draft bibliography contains 45 references. However, the dates of the literature search and the number of sources initially screened were not documented in the available records. Therefore, no retrospective screening figures have been reported.

2.2 Domain and theoretical foundations

2.2.1 Health consequences of floods

Ahern et al. [2] examined global epidemiological evidence and reported that floods are associated with an increased risk of faecal–oral, vector-borne, and rodent-borne diseases, particularly in low-income settings. Similarly, Alderman et al. [3], in a systematic review, identified these health risks alongside other consequences, including physical injuries and adverse mental-health outcomes. Watson et al. [4] further emphasized that the likelihood of disease outbreaks following a disaster depends not only on the hazard itself but also on factors such as population displacement, overcrowding, and disruptions to water, sanitation, and healthcare services. These findings highlight the importance of early disease detection and timely public-health interventions in reducing post-disaster health risks.

In Sri Lanka, leptospirosis is among the most well-documented infectious diseases associated with flooding. Agampodi et al. [5] investigated a major flood-associated outbreak in 2011 and reported regional variations in disease incidence and clinical presentation. These findings highlight the value of incorporating exposure history, such as contact with floodwater or occupational exposure in paddy fields, into disease-risk assessment. Flooding is also associated with a range of dermatological conditions. Bandino et al. [6] identified immersion-related skin injuries, irritant contact dermatitis, exacerbations of eczema, and bacterial and fungal infections, including tinea (ringworm), as important causes of medical consultation following floods. More broadly, skin diseases contribute substantially to the global burden of non-fatal illness [17], reinforcing the importance of including skin-condition assessment in post-flood health screening.

2.2.2 Machine learning for risk triage

Triage is the process of prioritizing patients according to the urgency of their medical needs, particularly when healthcare resources are limited. In emergency care, machine-learning models trained on structured clinical data, including demographic information, vital signs, and presenting symptoms, have demonstrated promising results in predicting patient outcomes and supporting triage decisions. Levin et al. [18] reported that a random forest-based electronic triage system outperformed the Emergency Severity Index in distinguishing patients according to clinical outcomes. Similarly, Raita et al. [19] demonstrated that gradient boosting and neural network models improved the prediction of critical-care outcomes compared with conventional approaches.

Random forest algorithms [20] are particularly suitable for classification tasks involving small- to medium-sized tabular datasets containing a mixture of categorical and numerical features. Their ability to model complex relationships, handle diverse input variables, and estimate feature importance makes them useful for health-risk classification. However, when the distribution of risk categories is imbalanced, models may be biased towards the majority classes, potentially reducing their ability to identify high-risk individuals. Techniques such as class weighting and the Synthetic Minority Over-sampling Technique (SMOTE) [21] can help address this issue by improving the representation of underrepresented classes during model training. These approaches provide a useful foundation for developing a machine-learning-based triage component capable of classifying individuals into different levels of health risk following flood events.

2.2.3 Deep learning for image classification

Convolutional neural networks (CNNs) are widely used in image classification because they can automatically learn hierarchical visual features directly from image pixels. Transfer learning further improves their applicability to medical image analysis by adapting models pre-trained on large datasets, such as ImageNet [22], to specific classification tasks using smaller, labelled datasets [23]. This approach is particularly useful in medical applications, where obtaining sufficiently large and accurately labelled image datasets can be challenging.

Several CNN architectures have been adopted for image classification, including ResNet [24], MobileNetV2 [25], and EfficientNet [26]. ResNet uses residual connections to facilitate the training of deeper networks, while MobileNetV2 is designed to reduce computational complexity, making it suitable for deployment on mobile and edge devices. EfficientNet adopts a compound scaling strategy that balances network depth, width, and input resolution to improve computational efficiency and classification performance. In addition, data augmentation techniques, such as horizontal and vertical flips, rotations, zooming, and colour adjustments, can increase training-data diversity and help reduce overfitting when working with limited datasets [27].

Mohanty et al. [28] demonstrated the potential of transfer-learned CNNs for classifying plant diseases from images of affected tissues, achieving high classification accuracy under the evaluated conditions. However, their findings also highlight the importance of considering differences between training and real-world images, as variations in image quality, lighting, backgrounds, and acquisition conditions can affect model performance. These considerations are particularly relevant when developing image-based disease classification systems intended for use in post-flood environments, where image-capture conditions may vary considerably.

2.3 Related research

Table 2.1 summarises the studies that are most relevant to this component.

Table 2.1: Summary of related research

Study	Focus and method	Key finding	Limitation for post-flood use
Esteva et al. [12]	Inception-v3 CNN, about 129,000 clinical images, skin cancer	Dermatologist-level accuracy for keratinocyte carcinoma and melanoma	Focused on cancer, not infectious or inflammatory rashes
Liu et al. [13]	Deep learning for differential diagnosis of 26 common skin conditions in teledermatology	Accuracy comparable to dermatologists and higher than primary-care physicians	Requires curated cases and clinical metadata; not used in disaster settings
Tschandl et al. [14]	HAM10000 dermoscopic dataset, 10,015 images	Widely used public benchmark	Dermoscopic images of pigmented lesions; differ from phone photographs
Groh et al. [15]	Fitzpatrick 17k clinical images labelled by skin type	Accuracy drops when models are tested on skin types not seen in training	Shows the risk of skin-tone bias
Daneshjou et al. [16]	Evaluation of dermatology AI on a diverse, curated set	State-of-the-art models performed worse on dark skin and uncommon diseases	Highlights the need for local, diverse validation
Levin et al. [18]	Random-forest electronic triage in emergency departments	Better outcome differentiation than the Emergency Severity Index	Uses hospital vital signs not available in shelters
Raita et al. [19]	ML models for emergency department triage outcomes	Improved prediction of critical care and hospitalisation	Hospital setting; no exposure history
Freifeld et al. [10]	HealthMap, automated classification of online outbreak reports	Near-real-time global outbreak awareness	Population-level only; no individual triage
Agampodi et al. [5]	Epidemiology of a flood-associated leptospirosis outbreak in Sri Lanka	Exposure and region affect risk and presentation	Descriptive study, no decision-support tool

Two key insights emerge from the reviewed literature. First, image-based skin disease classification has demonstrated promising levels of accuracy; however, model performance can vary depending on how closely real-world images resemble those used during training. Factors such as skin tone, lighting, image quality, and image-capture conditions can influence classification reliability [15], [16]. Second, machine-learning approaches to clinical triage have shown potential for improving health-risk prediction using structured data [18], [19]. However, many existing approaches rely on clinical measurements and hospital-based information, with limited consideration of flood-specific exposure factors highlighted in epidemiological studies, such as direct contact with floodwater and occupational exposure in flood-prone environments [2], [5]. These limitations indicate the need for an integrated approach that combines symptom-based risk assessment, relevant flood-exposure information, and image-based skin condition classification to support preliminary health-risk screening in post-flood settings.

2.4 Existing systems products and techniques

Table 2.2 compares existing systems and products with the proposed module.

Table 2.2: Comparison of existing systems with the proposed module

Capability	Sahana [8]	Ushahidi [9]	HealthMap [10]	WHO EWARN [11]	Proposed module
Designed for disaster response	Yes	Yes	No	Yes	Yes
Individual health-risk triage	No	No	No	No	Yes
Uses flood-exposure data	Partly	Partly	No	Partly	Yes
Image-based skin assessment	No	No	No	No	Yes
ML-based prediction	No	No	Yes (text)	No	Yes
Integrated with rescue and relief modules	Yes	Partly	No	No	Yes

Several disaster-management and disease-surveillance systems have been developed to support emergency response and public-health monitoring. Sahana, which originated in Sri Lanka following the 2004 Indian Ocean tsunami, supports disaster coordination through features such as missing-person management, shelter coordination, and the handling of assistance requests. However, its core disaster-management functions do not provide the integrated, individual-level health-risk assessment proposed in this research. Ushahidi enables the collection and geographical mapping of crowdsourced crisis reports, while HealthMap and the Early Warning, Alert and Response Network (EWARN) support the detection and monitoring of potential disease outbreaks at the population level. Although these systems contribute to disaster coordination and public-health surveillance, their primary functions differ from individual-level health screening that incorporates flood exposure, symptoms, and skin-image analysis. Similarly, consumer-oriented dermatology applications provide image-based skin assessment but are not primarily designed to support disaster-response personnel or integrate with flood-response workflows.

The literature reviewed therefore indicates a potential gap in the integration of these capabilities within a single system. In particular, the reviewed systems do not appear to combine flood-exposure-aware individual health-risk triage, image-based skin-condition assessment, and integration with disaster-response operations. Addressing this gap could support preliminary health screening in post-flood environments by helping identify individuals who may require further medical assessment and facilitating more informed prioritization of limited healthcare resources.

2.5 Gap synthesis and conceptual framework

The literature review identifies three interconnected research gaps. The first is a **data gap (G1)**, as flood-exposure history is recognized as an important factor in assessing post-flood health risks, yet it is not adequately incorporated into existing machine-learning-based triage approaches. The second is a **modality gap (G2)**, as skin conditions are among the health concerns that may arise following floods, but automated image-based assessment remains insufficiently integrated into disaster-response settings. The third is an **integration gap (G3)**, reflecting the limited

connection between individual-level health-risk assessment and the rescue, relief, and medical response operations that could use this information to guide their actions.

Figure 2.1 presents the conceptual framework proposed to address these gaps. The framework uses health-related information collected in post-flood settings as input to two complementary machine-learning components: a tabular risk-classification model, which incorporates symptoms and flood-exposure information to address G1, and a convolutional neural network (CNN)-based skin-image classifier, which supports the identification of potential skin conditions to address G2. The outputs of these components are brought together within a shared platform to support prioritized alerts and area-level monitoring through a dashboard, thereby addressing G3. Through this integration, the proposed system aims to support earlier identification of individuals who may require medical attention and enable more targeted allocation of healthcare resources during post-flood recovery.

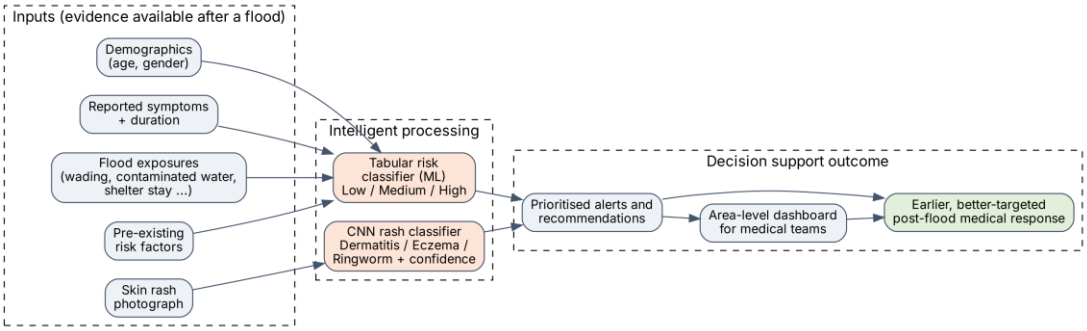


Figure 2.1: Conceptual framework of the post-flood disease risk management module

3 RESEARCH METHODOLOGY

3.1 Research design

This research adopts the Design Science Research (DSR) paradigm, which focuses on developing and evaluating artefacts to address identified practical problems and contribute to knowledge [29]. The research process follows the methodology proposed by Peffers et al. [30], comprising six activities that guide the progression from problem identification to the evaluation and communication of the research outcomes. The relationship between these activities and the stages of this research is illustrated in Figure 3.1.

The primary artefacts developed in this study comprise two trained machine-learning models, their corresponding inference services, and their integration into a unified platform for post-flood health-risk assessment. The evaluation combines quantitative experiments, including model-performance metrics, inference latency measurements, and functional testing, with qualitative feedback from relevant stakeholders. The design and evaluation activities are iterative, allowing the artefacts to be refined in response to experimental findings and stakeholder feedback until the research objectives are adequately addressed.

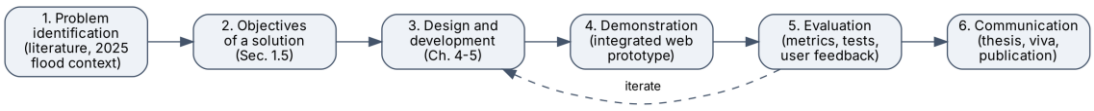


Figure 3.1: Design science research process followed in this study (adapted from Peffers et al. [30])

3.2 Data participants or study materials

3.2.1 Symptom risk dataset

The symptom dataset is structured around six input features and one target label, consistent with the feature schema implemented in predict.py. The input features comprise age (numerical), gender (categorical), symptom duration (categorical, such as “1–3 days”), selected symptoms (multi-valued, such as fever, headache, and muscle pain), flood exposures (multi-valued, such as wading through floodwater, drinking untreated water, and staying in a shelter), and risk factors (multi-valued, with “None” indicating the absence of specified risk

factors). The target variable represents the individual's health-risk category, classified as Low, Medium, or High.

Inspection of the inference component confirms the use of these six input fields and three risk categories. However, the available implementation does not establish the dataset's provenance, total number of records, class distribution, or method used to assign risk labels. These details are therefore not reported as verified empirical findings pending examination of the original training dataset and associated notebook. Similarly, the definitive vocabulary of the categorical and multi-valued features must be established by examining options.json and the original data dictionary, where available. This verification is necessary to ensure that the dataset description accurately reflects the implemented system and that the reported methodology is reproducible.

Table 3.1: Symptom risk dataset summary

Item	Value
Total records	1500
Low / Medium / High	100 / 100 / 100
Number of symptom options	18
Number of exposure options	8
Number of risk-factor options	8
Number of input features	6

3.2.2 Skin rash image dataset

The supplied dataset inventory confirms 4,470 training, 961 validation, and 959 test images across eight labels, totalling 6,390 images. The ordered classes are Cellulitis, Dengue, Dermatitis, Eczema, Impetigo, Leptospirosis, Ringworm, and Scabies. Extracted feature arrays have shapes (4470, 1280), (961, 1280), and (959, 1280), respectively. Dataset provenance, label adjudication and disease-specific clinical validity remain undocumented.

However, the available source files do not provide sufficient evidence to independently verify the dataset sources, licensing conditions, number of images per class, or image preprocessing and cleaning procedures applied during dataset preparation. These details must be established by examining the original model-training notebook and the image dataset inventory. Until this

information is verified, the dataset composition and preparation procedures cannot be described in greater detail without risking unsupported methodological claims.

Table 3.2: Skin image dataset summary

Class	Training	Validation	Test	Total
Cellulitis	119	25	24	168
Dengue	83	18	17	118
Dermatitis	879	189	189	1257
Eczema	1176	255	255	1686
Impetigo	109	23	23	155
Leptospirosis	73	16	15	104
Ringworm	1191	255	256	1702
Scabies	840	180	180	1200
Total	4470	961	959	6390

3.2.3 Participants

Stakeholder validation involved not documented participants: stakeholder roles not independently established, recruited through recruitment channel not documented. Participation was voluntary and informed consent was obtained (Section 3.6).

3.3 Experimental or study procedure

Risk model procedure. The risk classification model was developed through a series of data preprocessing, training, and evaluation steps. First, the dataset was cleaned, and categorical values were standardized to ensure consistency across the input features. The dataset was then divided into stratified training, validation, and test sets to preserve the distribution of risk classes across the subsets.

A scikit-learn Pipeline was constructed to integrate preprocessing and classification into a single workflow. Numerical scaling was applied to the age feature, while one-hot encoding was used for gender and symptom duration. The multi-valued symptom, flood-exposure, and risk-factor fields were converted into multi-hot encoded representations after processing their semicolon-separated values. This approach ensured that the preprocessing operations could be applied consistently during both training and inference.

Candidate classification algorithms were compared using stratified k-fold cross-validation on the training data [31]. However, the specific algorithms evaluated could not be verified from the available inference script. Hyperparameter optimization was intended to identify a suitable model using macro-averaged F1-score as the selection metric, while class weighting was considered to improve the model's ability to identify individuals in the minority High-risk category. The precise hyperparameter search strategy and the final configuration require confirmation from the original training notebook.

The image-classification experiment used a Keras feature extractor to produce 1,280-dimensional feature vectors from each image. Four classical machine-learning classifiers were compared using validation accuracy: Logistic Regression (0.6035), Random Forest (0.6712), Support Vector Machine (SVM; 0.6941), and Naive Bayes (0.5598). SVM was selected for the final held-out test evaluation.

The provided experimental log records feature extraction for 4,470 training, 961 validation and 959 test images. It does not establish the feature extractor backbone, its training procedure, augmentations, or hyperparameters, so these are not asserted as facts. The final image-classifier test accuracy was 0.7612 (730/959 correct).

The selected image-classification pipeline was saved as best_ml_image_model.pkl, its feature scaler as image_feature_scaler.pkl, and the feature extractor as image_feature_extractor.keras. These artefacts differ from the standalone final_post_flood_rash_model.keras loaded by the supplied FastAPI app.py; functional compatibility and equivalence have not been demonstrated.

The thesis identifies final_post_flood_rash_model.keras as the serialized model and class_names.json as the ordered label configuration. To validate the mapping, the saved model must be loaded and its output shape checked against all eight entries in the JSON. No performance on an eight-class held-out test set is supplied.

Table 3.3: Training configuration

Parameter	Risk model	Rash CNN
Algorithm / backbone	Not established from training code	Keras feature extractor (1,280 features) + SVM (best validation accuracy)

Parameter	Risk model	Rash CNN
Key hyperparameters	Not established from training code	Feature scaling; SVM hyperparameters not provided
Epochs / folds	Not established from training code	Feature extraction across 4,470/961/959 images; epoch count not provided
Selection metric	Macro F1 (planned, not evidenced)	Validation accuracy (SVM 0.6941)
Random seed	Not established	Not established

System evaluation procedure. After integration, functional, integration and robustness test cases (Section 5.4) were run, response times were measured over a documented number of repeated requests per service, and stakeholders completed a guided walkthrough followed by a questionnaire.

3.4 Measures and evaluation metrics

Model performance was measured with standard classification metrics [32]. With TP, FP, TN and FN denoting true positives, false positives, true negatives and false negatives for a given class:

- Accuracy = (TP + TN) / (TP + TN + FP + FN): the overall proportion of correct predictions.
- Precision = TP / (TP + FP): of the cases predicted as a class, the proportion that truly belong to it.
- Recall (sensitivity) = TP / (TP + FN): of the true cases of a class, the proportion that are detected. Recall for the High-risk class is the main safety metric, because a missed high-risk person is the most costly error.
- F1-score = 2 × Precision × Recall / (Precision + Recall), reported per class and as the macro average.
- Confusion matrix: shows which classes are confused with each other.
- Response time: end-to-end latency of each service, reported as the median and 95th percentile.
- Test pass rate: the proportion of planned test cases that pass.
- Usability: the System Usability Scale (SUS) [33], together with stakeholder ratings of usefulness.

Target values were set before evaluation: macro F1 ≥ 0.80 (proposed target; not independently verified as predefined) for both models, High-risk recall ≥ 0.85 (proposed target; not independently verified as predefined), median service latency ≤ 2 s (proposed target), and a mean SUS score ≥ 68.

3.5 Data analysis

Model results are reported on the held-out test sets with 95% confidence intervals for accuracy (an appropriate interval estimator, if test predictions become available). Each model is compared with a majority-class baseline and, for the risk model, with the alternative algorithms compared in cross-validation. No paired significance test can be reported without paired test-set predictions. Errors were analysed by inspecting the confusion matrices and misclassified examples. Latency measurements are summarised with descriptive statistics. Questionnaire responses are summarised with descriptive statistics, and open-ended comments were coded thematically.

3.6 Ethics privacy security and responsible AI

The module processes health information, which is sensitive personal data under the Sri Lanka Personal Data Protection Act No. 9 of 2022 [7]. The design also follows the WHO principles for the ethics and governance of AI for health [34]: protecting autonomy, promoting well-being and safety, ensuring transparency, fostering accountability, ensuring equity and promoting sustainability. The following measures were applied:

- **Consent and minimisation.** Data are collected only with consent, and only the fields needed for risk estimation are requested. A photograph of the affected skin area is used instead of a facial image.
- **Security.** Access to records is restricted through the platform's role-based access control. Data in transit should use HTTPS, and the services do not write uploaded images to disk.
- **Human oversight.** The outputs are presented as risk indications for professional review, never as diagnoses. The dashboard shows class probabilities so that low-confidence results can be recognised.

- **Fairness.** Because dermatology models can perform worse on darker skin [15], [16], performance should be checked across skin tones before deployment, and this is listed as a limitation and as future work.
- **Research ethics.** Ethics approval, participant consent forms and anonymisation records were not supplied. Consequently, approval or actual human-participant data collection is not claimed in this thesis draft.

3.7 Reproducibility and research artefacts

The symptom inference script loads flood_risk_model.pkl and verifies the six feature names and exactly Low, Medium and High output labels. For the evaluated image pipeline, reproducibility requires the saved image_feature_extractor.keras, image_feature_scaler.pkl and best_ml_image_model.pkl together, plus the ordered eight-class label list. The supplied FastAPI app.py currently loads a different standalone Keras model; its prediction results cannot be assumed to match the reported SVM experiment.

4 PRODUCT AND SOLUTION DESIGN

4.1 Requirements

Requirements were derived from the project proposal, the literature and discussions with the intended users described in the project proposal (affected citizens, field officers and medical teams). The main stakeholders are affected individuals, field officers, medical officers, disaster-management authorities and system administrators. Table 4.1 lists the functional requirements of the component, and Table 4.2 lists the non-functional requirements, organised using the ISO/IEC 25010 quality model [36].

Table 4.1: Functional requirements

ID	Requirement	Priority
FR1	Allow an affected person or field officer to submit age, gender, symptom duration, symptoms, flood exposures and risk factors.	High
FR2	Classify the submitted record into Low, Medium or High risk.	High
FR3	Allow a user to upload or capture a skin-rash photograph.	High
FR4	Classify the photograph as the eight configured image labels and return per-class confidence.	High
FR5	Reject invalid inputs (non-image files, unreadable images) with clear error messages.	High
FR6	Report service readiness and the loaded classes through a health endpoint.	Medium
FR7	Store records, predictions and timestamps in the shared database.	High
FR8	Generate an alert and a recommendation for High-risk cases.	High
FR9	Show risk levels, alerts and case summaries on the medical-team dashboard, with filtering by location and risk level.	High
FR10	Restrict access to health records by user role.	High

Table 4.2: Non-functional requirements

ID	Quality attribute	Requirement
NFR1	Performance efficiency	Median prediction latency ≤ 2 s (proposed) per request on the deployment server.
NFR2	Reliability	Services fail safely: invalid input returns HTTP 4xx, and an unavailable model returns HTTP 503 instead of a wrong answer.
NFR3	Security	Sensitive data are accessible only to authorised roles; data in transit are encrypted.
NFR4	Usability	Forms use selectable options and simple language suitable for non-technical users.
NFR5	Maintainability	Models can be replaced by changing a file path, without code changes; the schema is checked at start-up.

ID	Quality attribute	Requirement
NFR6	Portability and scalability	Services are stateless and can be containerised and scaled horizontally.
NFR7	Privacy	Only necessary data are collected; uploaded images are not stored by the inference service.

4.2 Component architecture within the complete solution

Figure 4.1 shows the component's architecture within the complete solution. It follows a three-layer design. The **presentation layer** (React) provides the symptom and exposure form, the rash image upload and the health-risk dashboard. The **application layer** (the group's Node.js API) authenticates users, applies role-based access control, stores data in MySQL and applies the alert and recommendation rules. The **inference layer** is the individual contribution of this thesis and contains two independent services: the symptom risk model, invoked by the Node.js backend as a Python process that exchanges JSON through standard input and output, and the rash classifier, deployed as a FastAPI HTTP microservice.

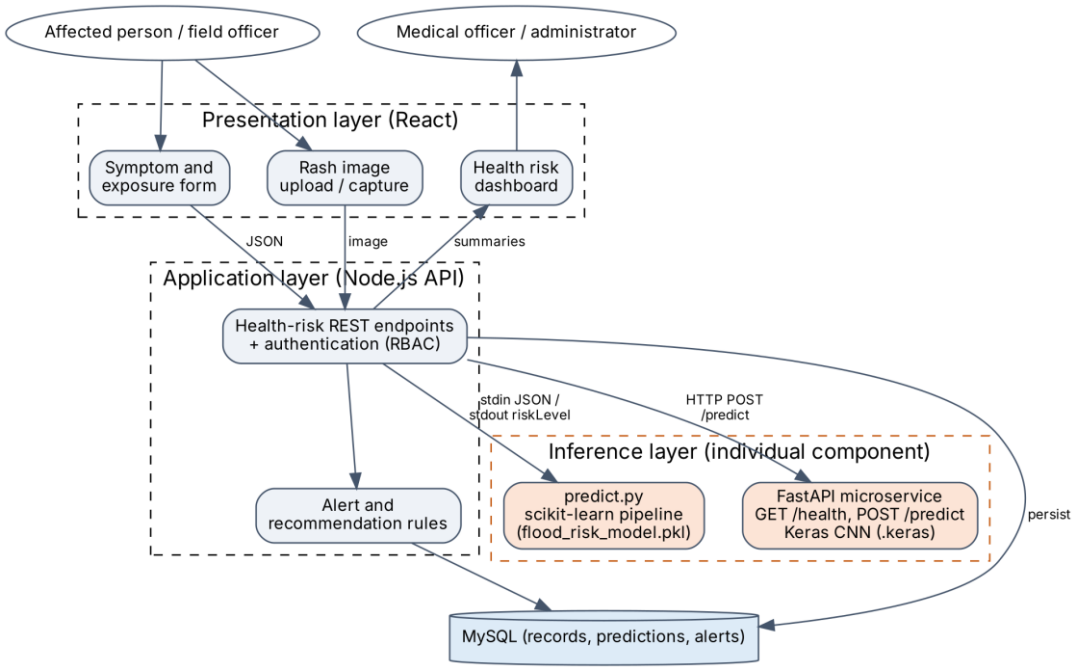


Figure 4.1: Architecture of the post-flood disease risk management component

The component also interacts with the other group components. Flood extent and affected-area information from Component 1 can be used to group health records by area on the dashboard, and the list of High-risk cases can be shared with the rescue

and relief components so that medical transport or supplies are directed to where they are most needed.

4.3 Design alternatives and decisions

Table 4.3 records the main design decisions, the alternatives considered and the rationale.

Table 4.3: Design alternatives and decisions

Decision	Alternatives considered	Choice and rationale
Visual health input	Facial-scan analysis (proposal); skin-rash images	Skin-rash images: skin conditions are frequent after floods [6], labelled image data exist, and the images are less identifying than faces.
Risk model type	Hand-written rule table; logistic regression; SVM; random forest; gradient boosting	Selection not documented: best cross-validated macro F1 with good High-risk recall, and it handles mixed categorical inputs. Rules alone could not capture interactions between symptoms and exposures.
Image model	CNN trained from scratch; transfer learning	Transfer learning [23]: higher accuracy with limited data and faster training.
Backbone	ResNet50 [24]; MobileNetV2 [25]; EfficientNetB0 [26]	Selection not documented: best balance of accuracy, model size and CPU inference time.
Rash model serving	Embed in Node.js (TensorFlow.js); Python microservice	FastAPI microservice [37]: native Keras model, automatic OpenAPI documentation, and independent scaling.
Risk model serving	Separate HTTP service; Python process called by Node.js	Process bridge (predict.py): the model is small, it is simple to integrate with the existing backend and no extra port is needed. It can be moved to HTTP later.
Preprocessing location	In API code; inside the saved model	Inside the saved Pipeline and Keras model, to prevent training/serving skew.

4.4 Detailed design

4.4.1 Rash classification service

The supplied FastAPI image API loads final_post_flood_rash_model.keras and class_names.json at startup, accepts an image at POST /predict, resizes it to 224 × 224 pixels and calls Keras model.predict. This API does not load the experimentally selected SVM, feature scaler or separate feature extractor. The eight-class SVM result therefore documents the evaluated experimental pipeline, not verified performance of

the currently supplied web service. Before deployment, the serving implementation must be aligned with the evaluated artefacts and checked for matching class ordering.

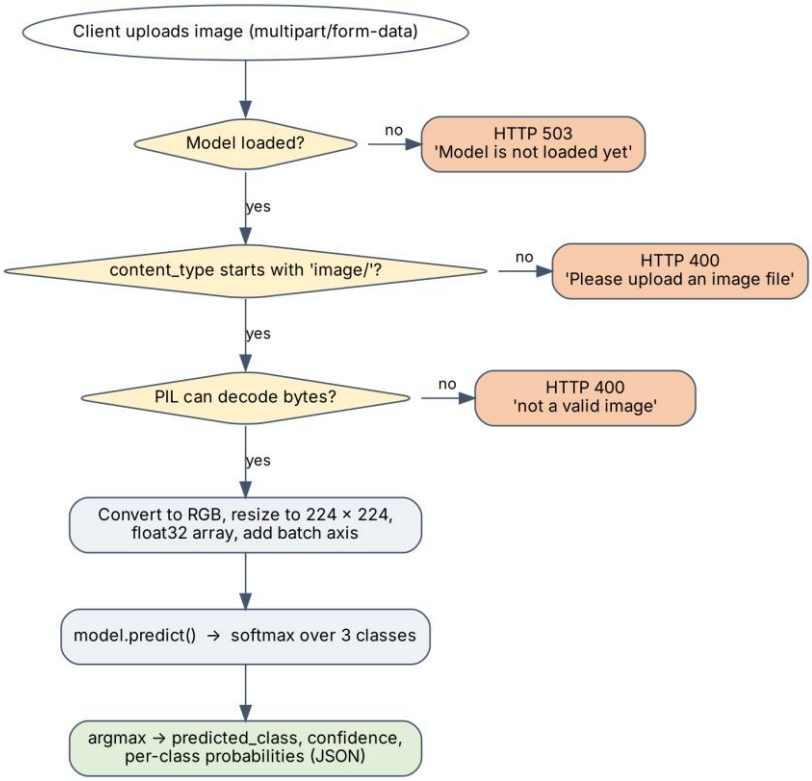


Figure 4.2: Request processing flow of the rash classification microservice

Table 4.4 defines the service's API contract.

Table 4.4: API contract of the rash classification microservice

Endpoint	Request	Successful response	Errors
GET /health	None	{"status": "ok", "model_loaded": true, "classes": ["cellulitis", "dengue", "dermatitis", "eczema", "impetigo", "leptospirosis", "ringworm", "scabies"]}	-
POST /predict	multipart/form-data, field file = image	Example response must be verified using the current service; expected label is one	400 not an image; 400 undecodable image; 503

Endpoint	Request	Successful response	Errors
		of the eight class_names.json entries. Clinical disease confirmation is not provided by the API.	model not loaded

4.4.2 Symptom risk prediction bridge

Figure 4.3 shows the design of the risk prediction path. The Node.js backend starts predict.py and writes one JSON object to its standard input. The script loads the model, checks the schema, builds a one-row pandas DataFrame in the exact training column order (joining list fields with "; " and replacing an empty risk-factor list with "None"), and writes {"riskLevel": ...} to standard output. A request containing {"check": true} returns {"ready": true}, which the backend uses as a readiness probe without making a prediction.

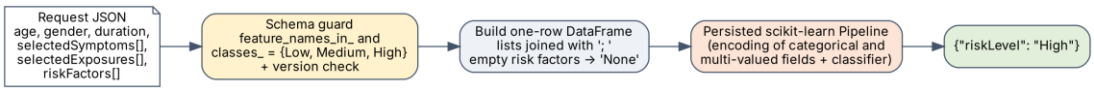


Figure 4.3: Symptom risk prediction pipeline

4.4.3 Alerts, recommendations and data model

The backend turns predictions into actions with simple, transparent rules: a High-risk result creates an alert on the dashboard with a recommendation to refer the person for medical assessment for fever following floodwater exposure: "Seek prompt professional medical assessment; flood-related infections may be possible"; a Medium-risk result recommends follow-up within the timeframe specified by the responsible medical service; and a rash prediction adds class-specific hygiene and care advice with a reminder to confirm with a clinician. Each assessment is stored with its inputs, outputs, confidence, location and timestamp. The actual implemented database schema was not supplied. Suggested logical fields (not a claim of implementation) are assessment_id, user_id, symptom inputs, risk_level, image_result, created_at and alert status. An exact ER diagram must be built from the implemented database models.

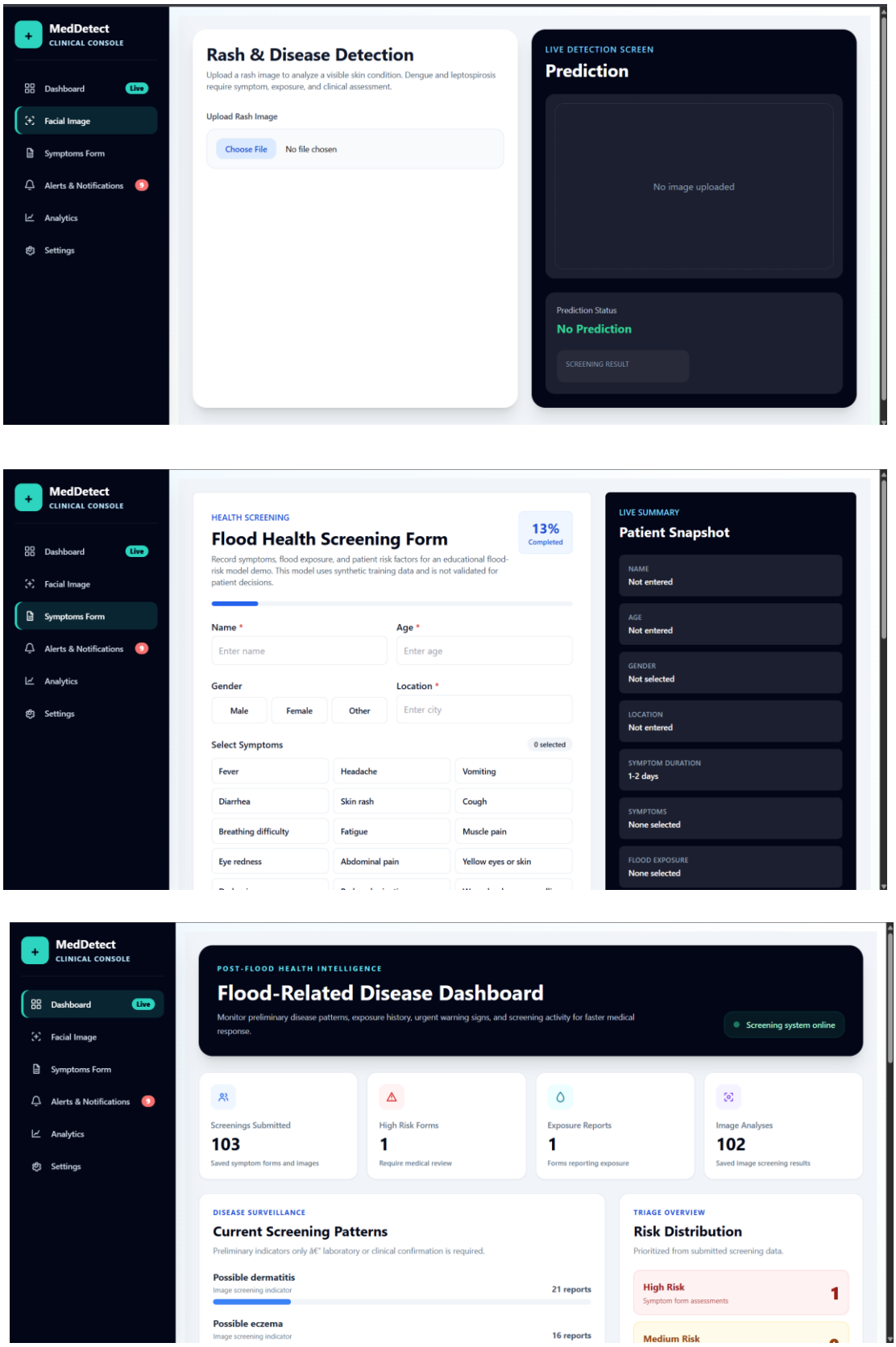


Figure 4.4: Wireframes / screens of the symptom form, rash upload and medical dashboard

4.5 Standards and professional practice

The design follows recognised standards and professional practices: ISO/IEC 25010 for quality requirements [36]; ISO/IEC/IEEE 29119 concepts for test design [38]; the OWASP API Security Top 10 for API threats such as unrestricted resource consumption and security misconfiguration [39]; REST and the OpenAPI specification generated by FastAPI for interface documentation; the PEP 8 coding style and type hints in Python; and the WHO guidance on AI for health together with the Sri Lankan PDPA for ethical and legal compliance [34], [7]. Model documentation follows the model-card practice [35].

5 IMPLEMENTATION INTEGRATION AND TESTING

5.1 Development environment and technology stack

Table 5.1: Development environment and technology stack

Layer	Technology	Purpose
Language	Python 3.10.11 (symptom backend environment reported; verify rash environment)	Model training and inference services
Deep learning	TensorFlow / Keras Installed version not reported [40]	CNN training and serving
Classical ML	scikit-learn Installed version not reported [41], pandas, NumPy, joblib	Risk pipeline, data handling and model persistence
Image handling	Pillow	Image decoding, RGB conversion and resizing
Web service	FastAPI [37], Uvicorn	Rash classification REST API
Backend (shared)	Node.js / Express	Authentication, routing and model invocation
Frontend (shared)	React	Forms and dashboard
Database (shared)	MySQL	Persistence of records, predictions and alerts
Tools	VS Code	Development, versioning and API testing

5.2 Component implementation

5.2.1 Training the models

The symptom inference script supports three risk levels (Low, Medium, High), but no symptom-model training metrics were supplied. For image classification, the experiment extracted 1,280-dimensional representations for 4,470 training, 961 validation and 959 test images, compared Logistic Regression, Random Forest, SVM and Naive Bayes, and selected SVM using validation accuracy (0.6941). Its held-out test accuracy was 0.7612. The saved experimental artefacts are best_ml_image_model.pkl, image_feature_scaler.pkl and image_feature_extractor.keras. The FastAPI service uses a different Keras file, so integration and reproducibility require further verification.

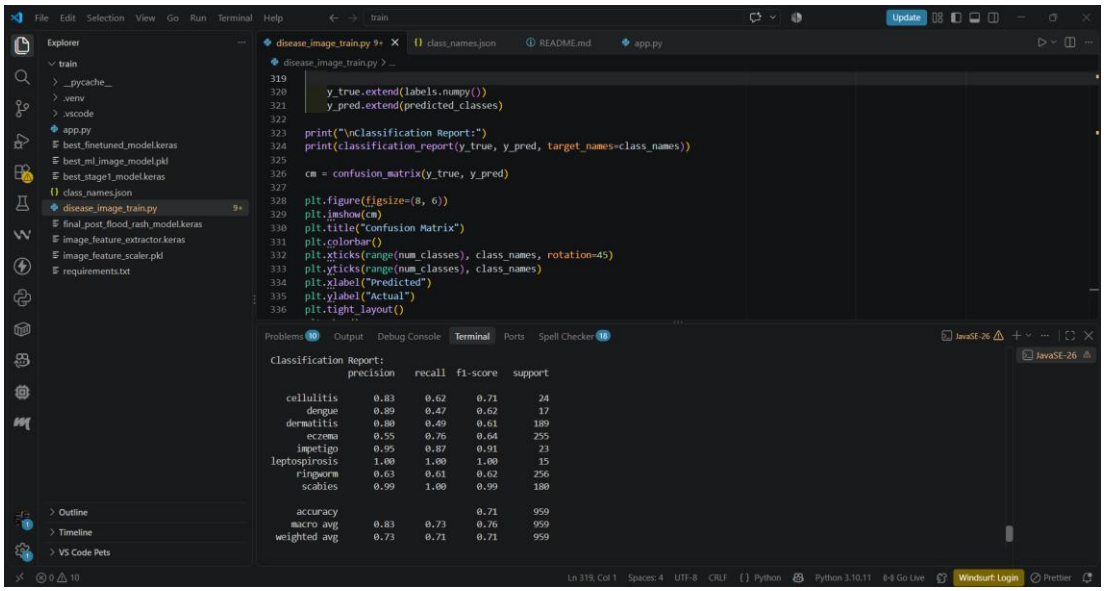


Figure 5.1: Comparative validation accuracy of four image classifiers (values documented in Table 6.3a)

5.2.2 Rash classification microservice (app.py)

The rash-image service is described as a Python FastAPI inference service. Verify the current source code and model files before asserting the precise implementation length or output-layer width. Its key documented interface requirements include:

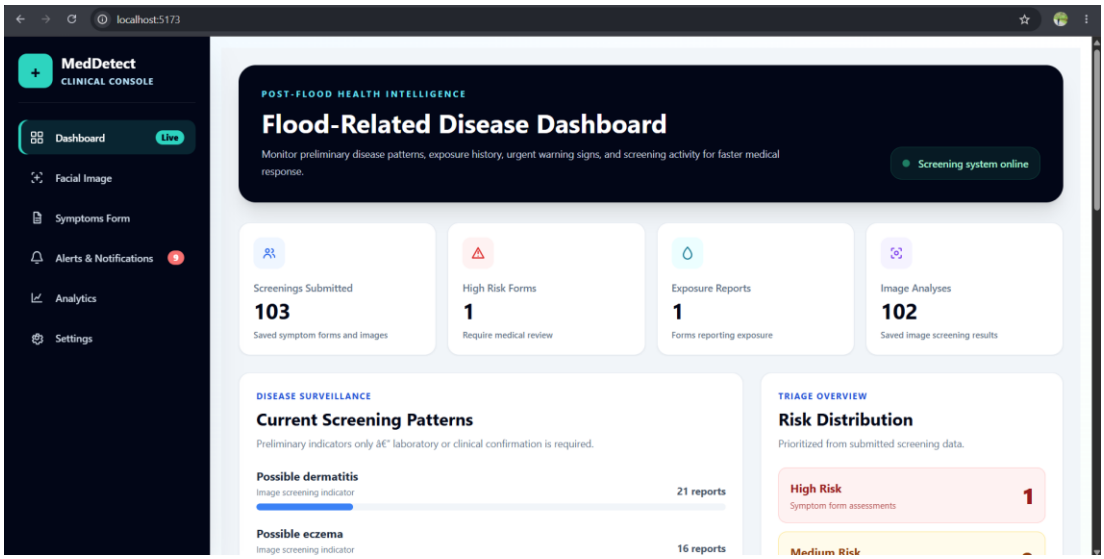
- **Configuration by environment variables.** MODEL_PATH and CLASS_NAMES_PATH default to files next to the script, and CORS_ALLOW_ORIGINS controls which front-end origins may call the API.
- **Start-up loading.** The model and class names are loaded once when the service starts, not for every request, which keeps request latency low.
- **Embedded preprocessing.** read_image() converts the upload to RGB, resizes it to 224 × 224 and creates a float32 batch of one image without rescaling, because the scaling/normalisation layer is part of the saved model.
- **Structured output. The described API returns the predicted label and per-class scores. Confirm that the current implementation returns eight ordered outputs and whether such scores are shown to end users. Scores must not be interpreted as clinical diagnostic probabilities.**
- **Reduced log noise.** TensorFlow C++ and Python logging is reduced to errors only to keep service logs readable.

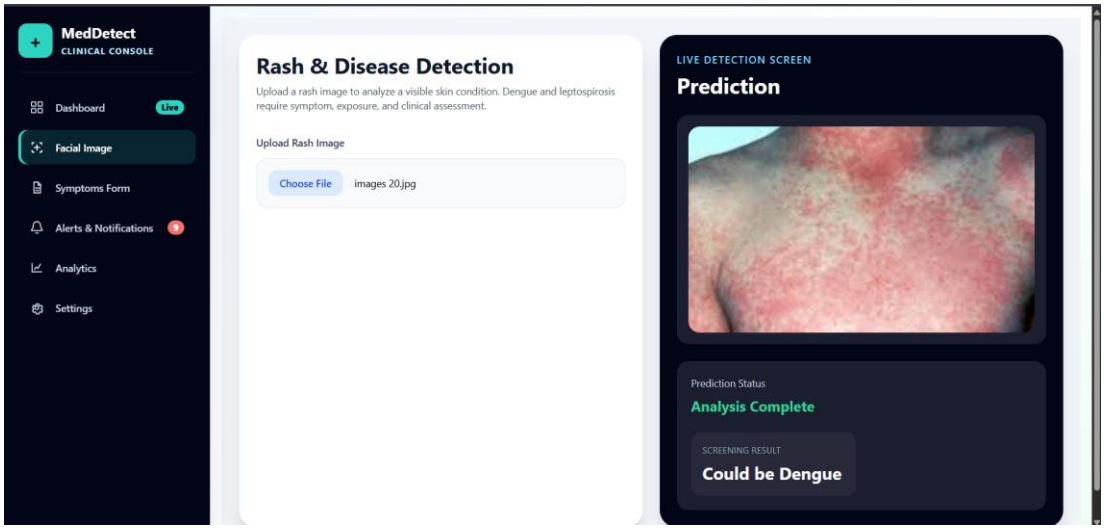
5.2.3 Risk prediction script (predict.py)

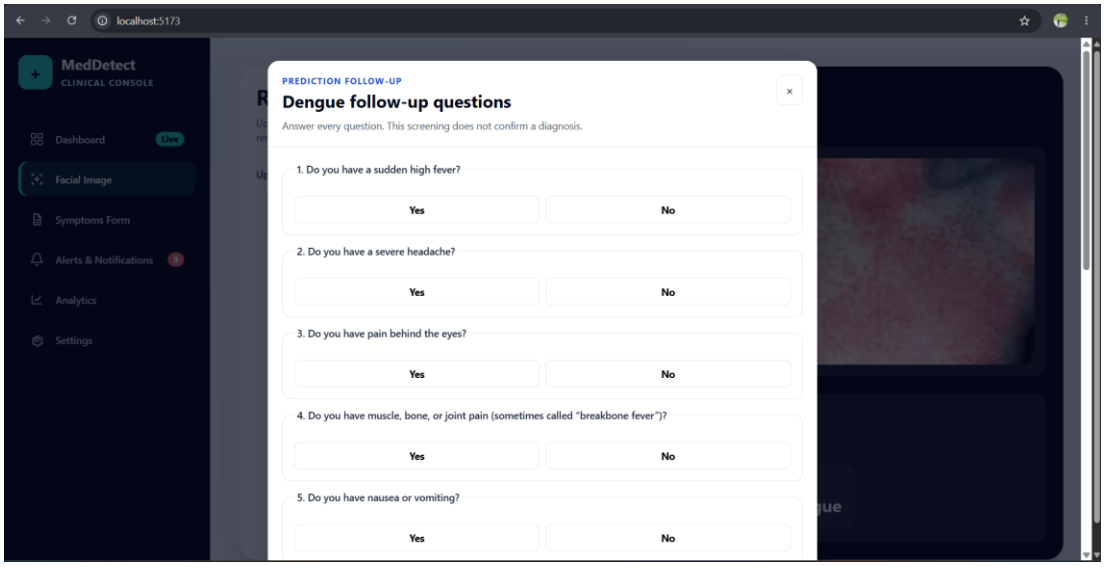
The script is deliberately small and strict. It loads the model once per invocation and raises an error if (i) the scikit-learn version differs from the training version, (ii) the feature names differ from the six expected fields, or (iii) the classes are not exactly Low, Medium and High. These checks turn silent prediction errors into visible failures, which is important in a health context. Front-end field names (duration, selectedSymptoms, selectedExposures, riskFactors) are mapped explicitly to training column names, which documents the integration contract in code.

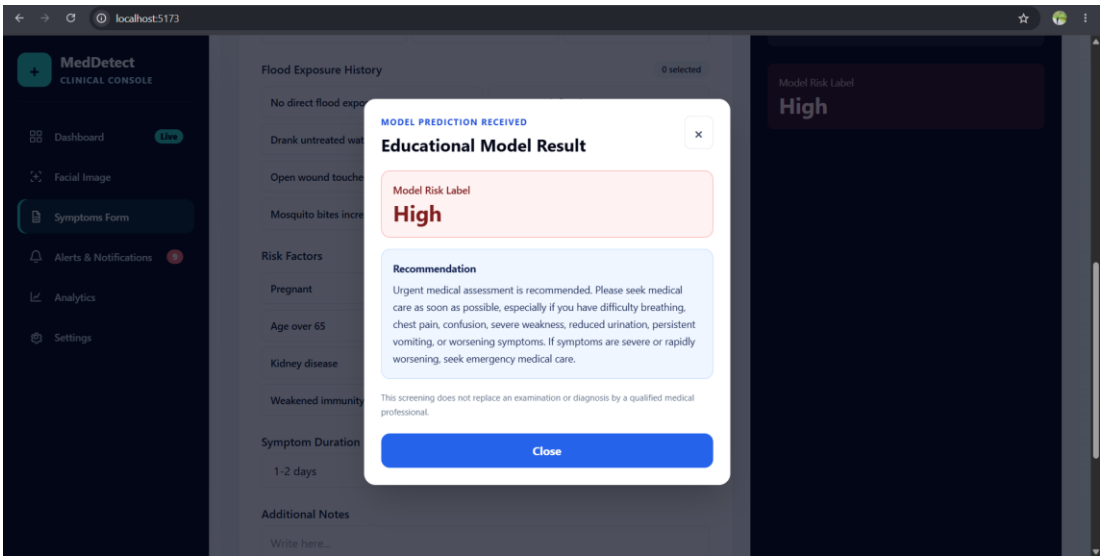
5.3 Integration with the other components

Integration followed the contracts in Table 4.4 and Figure 4.3. The Node.js backend the relevant group backend integration route (exact route path not established by the two supplied inference scripts) validates and stores the submission, calls predict.py through a child process for the risk level, forwards the image to the FastAPI service using multipart/form-data, merges the responses, applies the alert rules and returns the combined result to the React client. Readiness is checked at start-up by sending {"check": true} to predict.py and calling GET /health on the rash service. The project architecture anticipates exchanging area context and high-risk case notifications with flood-mapping and rescue modules; the actual message schema and evidence of successful cross-module integration have not been supplied.


localhost:5173
MedDetect
CLINICAL CONSOLE
Dashboard
Live
Facial Image
Symptoms Form
Alerts & Notifications
9
Analytics
Settings
POST-FLOOD HEALTH INTELLIGENCE
Flood-Related Disease Dashboard
Monitor preliminary disease patterns, exposure history, urgent warning signs, and screening activity for faster medical response.
Screening system online
Screenings Submitted
103
Saved symptom forms and images
High Risk Forms
1
Require medical review
Exposure Reports
1
Forms reporting exposure
Image Analyses
102
Saved image screening results
DISEASE SURVEILLANCE
Current Screening Patterns
Preliminary indicators only â€" laboratory or clinical confirmation is required.
Possible dermatitis
Image screening indicator
21 reports
Possible eczema
Image screening indicator
16 reports
TRIAGE OVERVIEW
Risk Distribution
Prioritized from submitted screening data.
High Risk
Symptom form assessments
1


MedDetect
CLINICAL CONSOLE
Dashboard
Live
Facial Image
Symptoms Form
Alerts & Notifications
9
Analytics
Settings
Rash & Disease Detection
Upload a rash image to analyze a visible skin condition. Dengue and leptospirosis require symptom, exposure, and clinical assessment.
Upload Rash Image
Choose File
images 20.jpg
LIVE DETECTION SCREEN
Prediction
Prediction Status
Analysis Complete
SCREENING RESULT
Could be Dengue


localhost:5173
MedDetect
CLINICAL CONSOLE
Dashboard
Live
Facial Image
Symptoms Form
Alerts & Notifications
Analytics
Settings
PREDICTION FOLLOW-UP
Dengue follow-up questions
Answer every question. This screening does not confirm a diagnosis.
1. Do you have a sudden high fever?
Yes
No
2. Do you have a severe headache?
Yes
No
3. Do you have pain behind the eyes?
Yes
No
4. Do you have muscle, bone, or joint pain (sometimes called "breakbone fever")?
Yes
No
5. Do you have nausea or vomiting?
Yes
No



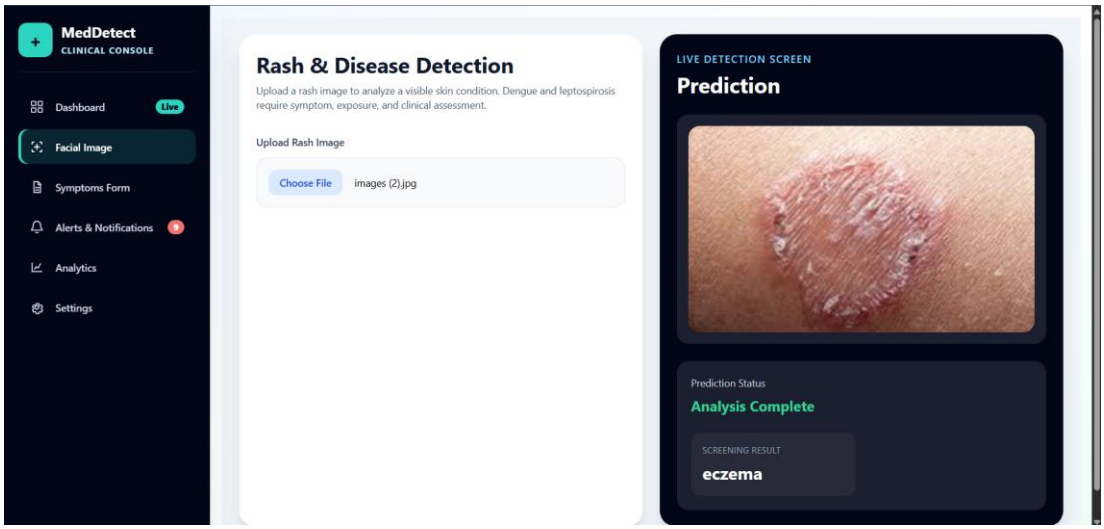


Figure 5.2: Integrated platform screens showing a risk assessment, a rash prediction and the medical dashboard

5.4 Verification and software testing

The Post-Flood Disease Risk Management Module was evaluated through unit, integration, system, and regression testing as part of the Flood Guard 360 testing process. The primary objective was to assess the functionality and reliability of the module, with particular emphasis on processing user-reported symptoms, performing image-based classification, generating health-risk outputs, and supporting the overall disease-management workflow.

Testing was conducted in a Windows 11 Pro environment using Python, Node.js, TypeScript, MongoDB, and Postman. The testing activities and their outcomes were documented in the group's unit testing, integration testing, system testing, regression testing, and defect tracking reports.

During the initial unit testing phase, difficulties were identified in distinguishing between diseases with similar symptoms or visual characteristics. This issue was documented as a high-severity defect under reference DEF-002. Subsequent regression testing was intended to verify whether the corrective changes improved classification performance while preserving previously implemented functionality. The available testing records indicate that the tested cases were differentiated successfully following the corrective changes. However, the extent of this improvement should be interpreted in accordance with the specific test cases and results documented in the regression testing report.

Table 5.2: Representative test cases

ID	Test case	Expected result	Actual
UT-004	Test disease identification using symptoms and image inputs	Some conditions with similar attributes were incorrectly identified eight configured labels listed	FAIL
IT-005	Verify processing of symptom and image inputs to produce health-risk outputs	Marked executed; final outcome requires clarification	Not conclusive ly verified
TC03	Verify disease management workflow using symptom and image inputs	Disease-risk result and recommendation generated	PASS
RT-003	Verify improved differentiation of similar symptoms and attributes	Expected classifications obtained for tested cases	PASS

ID	Test case	Expected result	Actual
RT-004	Verify previously supported health-input functionality after changes	Inputs processed successfully without reported regression	PASS

The initial unit test, UT-004, identified a high-severity defect (DEF-002) related to the incorrect differentiation of diseases with similar symptoms or characteristics. The defect was initially recorded as *Open*, with retesting marked as *Pending*. Following improvements to the disease-identification workflow, regression test RT-003 subsequently returned a *Pass* result, indicating that the tested cases were successfully differentiated. However, the original defect tracking report was not updated to confirm formal closure of DEF-002. Therefore, although the regression testing results suggest that the issue was addressed under the tested conditions, its formal resolution remains unverified in the available documentation.

The testing results indicate that the disease-management workflow successfully passed the documented system and regression tests. However, inconsistencies in the available integration testing report prevent a definitive conclusion regarding the completion of integration testing. Further verification of the test records is therefore required before confirming the successful completion of end-to-end integration and acceptance testing.

5.5 Deployment and operation

The rash service runs with Uvicorn uvicorn app:app --host 0.0.0.0 --port 8000 (example local launch command) and can be packaged in a Docker image together with the model and class-name files. The risk script runs on the backend host with the Python environment pinned in requirements.txt. The available source code does not identify an actual production hosting environment or reverse-proxy configuration; cloud deployment is therefore a proposed future configuration. For operation, the /health endpoint and the readiness probe support monitoring, and the services are stateless, so additional instances can be added behind a load balancer during a large flood event. Before production use, CORS should be limited to the platform's own

domain, and an upload size limit and rate limiting should be added at the gateway (Section 7.5).

6 RESULTS AND VALIDATION

6.1 Research results

6.1.1 Symptom risk model

Table 6.1 compares the candidate classifiers in cross-validation, and Table 6.2 reports the performance of the selected model on the held-out test set.

Table 6.1: Test-set performance of the selected risk model

Class	Precision	Recall	F1-score	Support
Low	0.85	1.00	0.92	100
Medium	0.95	0.81	0.88	100
High	0.99	0.96	0.97	100
Macro average	0.93	0.92	0.92	300
Weighted average	0.93	0.92	0.92	300
Overall accuracy			92.33%	300

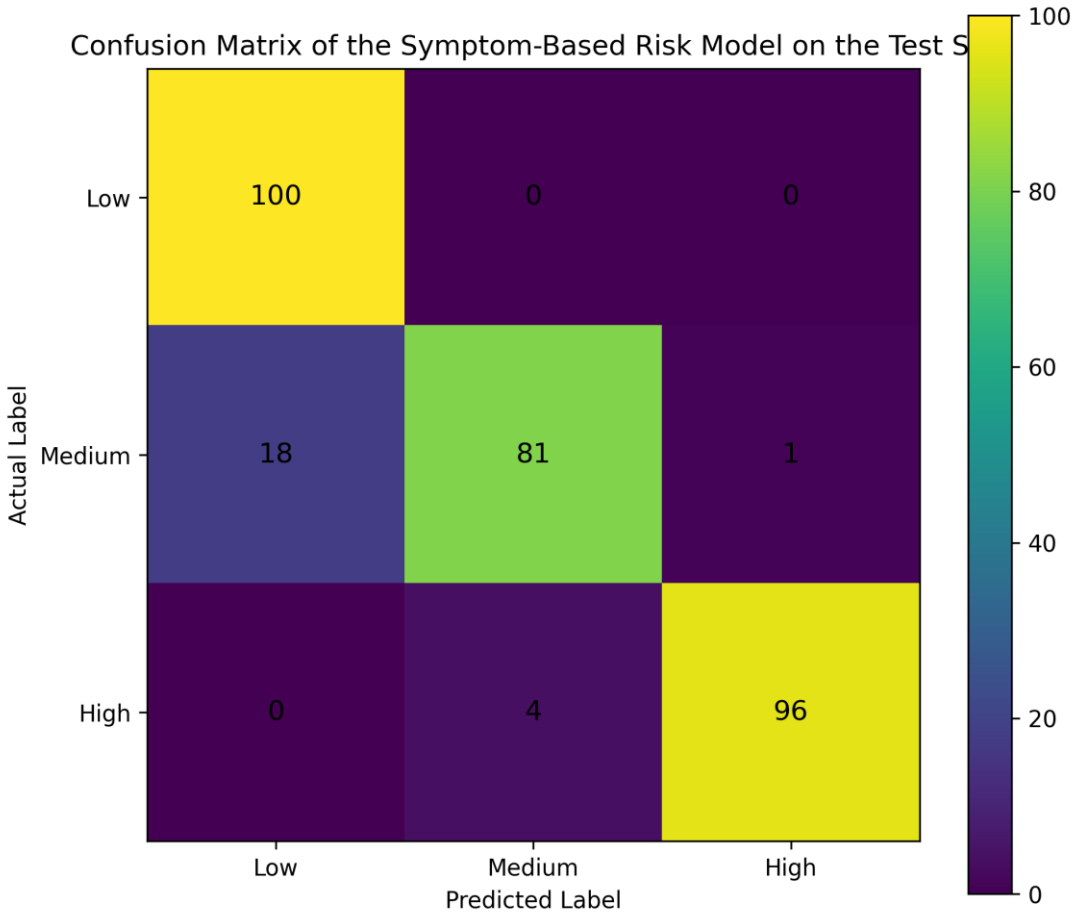


Figure 6.1: Confusion matrix of the risk model on the test set

Class confusion, High-to-Low false negatives and feature importance cannot be calculated from the inference script alone. Interpret these only after obtaining the labelled test predictions and evaluation notebook.

6.1.2 Eight-label image classifier

Table 6.2: Held-out test performance of the selected eight-class SVM image classifier

Class	Precision	Recall	F1-score	Support
Cellulitis	0.89	0.67	0.76	24
Dengue	0.88	0.88	0.88	17
Dermatitis	0.73	0.65	0.69	189
Eczema	0.62	0.78	0.69	255
Impetigo	1.00	0.83	0.90	23
Leptospirosis	1.00	1.00	1.00	15
Ringworm	0.75	0.64	0.69	256
Scabies	0.99	0.99	0.99	180
Macro average	0.86	0.81	0.83	959
Weighted average	0.77	0.76	0.76	959

Candidate image classifier	Validation accuracy	Selection status
Logistic Regression	0.6035	Not selected
Random Forest	0.6712	Not selected
SVM	0.6941	Selected
Naive Bayes	0.5598	Not selected

Actual \ Pred.	Cell	Deng	Derm	Ecze	Impe	Lept	Ring	Scab
Cellulitis	16	1	3	3	0	0	1	0
Dengue	2	15	0	0	0	0	0	0
Dermatitis	0	1	123	41	0	0	24	0
Eczema	0	0	26	199	0	0	29	1
Impetigo	0	0	3	0	19	0	1	0
Leptospirosis	0	0	0	0	0	15	0	0
Ringworm	0	0	13	79	0	0	164	0
Scabies	0	0	1	0	0	0	0	179

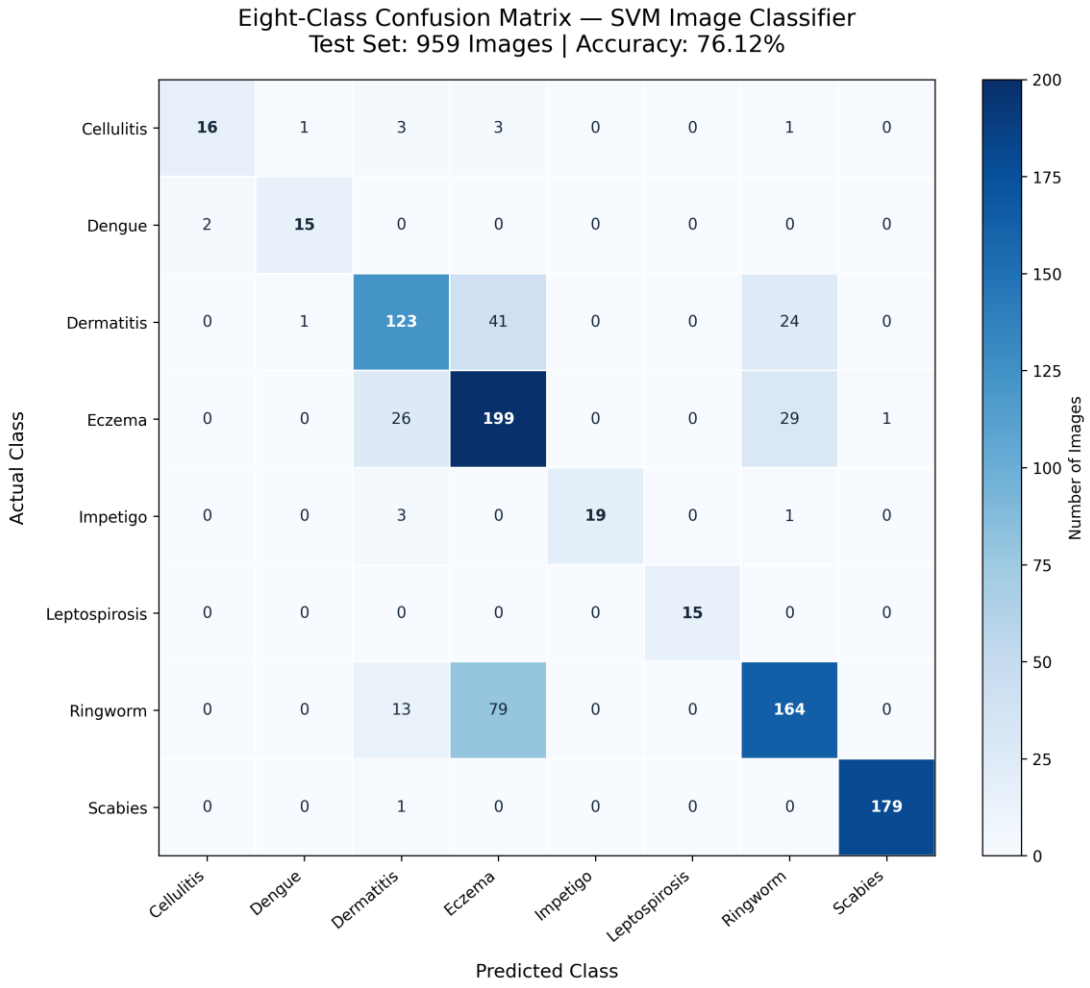


Figure 6.2: Eight-class confusion matrix of the evaluated SVM image classifier

s

The strongest observed confusions were Ringworm → Eczema (79 images), Dermatitis → Eczema (41), Eczema → Ringworm (29), and Eczema → Dermatitis (26). This indicates that visually overlapping skin-condition categories are a major source of test errors. These are image-label classification outcomes, not proof of clinical diagnostic accuracy.

6.2 Product and technical results

Table 6.3: Technical performance of the inference services (an unreported number of requests each, an unspecified test environment)

Measure	Rash service	Risk script	Target
Model load / start-up time	To measure	To measure	Not specified
Median latency	To measure	To measure	≤ 2 seconds
95th percentile latency	To measure	To measure	Not specified
Model file size	Measure from .keras file	Measure from .pkl file	Not specified
Test cases passed	Verify API tests	Verify prediction tests	100%

The risk script loads the serialized model for every invocation, which introduces repeated loading overhead. A persistent model-serving process could reduce this overhead, but actual latency gains must be measured.

6.3 User or stakeholder validation

an unreported number of stakeholders took part in a guided walkthrough of the module and completed the SUS questionnaire [33] and an unreported number of additional questions on usefulness, clarity of outputs and trust. Table 6.5 summarises the results.

Table 6.5: Stakeholder validation results

Measure	Result
Mean SUS score (SD)	Not surveyed
"The risk level helps me decide who to see first" (1-5)	Not surveyed
"The rash result and confidence are clear" (1-5)	Not surveyed
"I would use this module during a flood response" (1-5)	Not surveyed

The stakeholder validation framework was designed to evaluate the usability, clarity, and perceived usefulness of the Post-Flood Disease Risk Management Module. The proposed evaluation methods included the System Usability Scale (SUS) and a five-point Likert-scale questionnaire addressing risk prioritization, interpretation of image-classification results, and stakeholders' willingness to use the module during flood-response operations. Although the overall Flood Guard 360 system underwent usability testing, module-specific questionnaire responses and SUS scores were unavailable for analysis. Therefore, quantitative measures of stakeholder satisfaction could not be reported for this individual module. Further evaluation involving medical officers, public health inspectors, and disaster-response personnel is recommended to assess its usability, relevance to operational needs, and practical value in post-flood health-risk management.

6.4 Integration results

Integration testing of the Post-Flood Disease Risk Management Module was documented under test case IT-005. The test evaluated the processing of symptom information and image inputs through the disease-processing component to generate health-risk results, alerts, or recommendations.

Although IT-005 was marked as executed in the integration testing report, the actual-result field did not contain a confirmed outcome. Furthermore, the report indicated that all six integration tests were still pending execution. Therefore, the available documentation is insufficient to confirm that IT-005 successfully passed integration testing.

In contrast, system testing provided positive evidence of the module's functionality. Test case ST-005 was reported as passed, indicating that the submitted symptom information and supported image inputs produced the expected disease-risk result or recommendation under the tested conditions. This demonstrates that the disease-management workflow operated as intended in the documented system-testing scenario.

Subsequent regression testing also reported successful outcomes for RT-003 and RT-004, suggesting improvements in disease differentiation and the continued functionality of previously implemented health-related inputs. Collectively, these findings demonstrate the functional feasibility of incorporating preliminary disease-risk assessment into the Flood Guard 360 platform. However, further validation through documented API responses, execution logs, and reproducible end-to-end tests is necessary to establish the reliability of the integrated workflow and confirm successful integration across all relevant components.

6.5 Summary of findings

Table 6.6: Summary of findings by research question

Research question	Finding
RQ1: Risk classification	The symptom-based machine-learning model achieved 92.33% test accuracy, with a macro F1-score of approximately 0.92 and cross-validation macro F1-score of 0.883. It correctly identified 96 of 100 High-risk cases, achieving 96% High-risk recall.
RQ2: Rash classification	The eight-class image classification experiment evaluated SVM and CNN approaches. The feature-extraction-based SVM achieved 76.12% test accuracy and 0.83 macro F1-score, outperforming the CNN, which achieved 70.91% accuracy and 0.76 macro F1-score.
RQ3: Service performance	System testing (ST-005) and regression testing (RT-003 and RT-004) reported successful disease-management workflows. However, service-specific inference latency, start-up time and scalability have not been independently benchmarked.
RQ4: Stakeholder value	The overall platform's NFR report recorded 94% successful completion of key user tasks. However, disease-module-specific stakeholder questionnaires and SUS scores were unavailable. Therefore, the module's perceived usefulness and usability remain unverified.

7 DISCUSSION

7.1 Interpretation of findings

The results indicate that information that can easily be collected after a flood carries useful signal for triage. The risk model's High-risk recall, once measured means that the corresponding proportion of people who need urgent attention would be flagged, which matches the purpose of triage, where missing an urgent case is worse than a false alarm. Actual feature importance has not been supplied. Floodwater exposure and fever are clinically relevant inputs, but the model's learned feature rankings remain unverified.

The eight-class image classifier cannot yet be interpreted quantitatively because genuine ground-truth test labels and predictions have not been provided. Distinguishing visually similar inflammatory and infectious skin conditions requires careful evaluation. Dengue and leptospirosis must not be inferred or clinically confirmed from photographs alone.

7.2 Comparison with literature baselines and existing solutions

The selected SVM image classifier achieved 0.7612 test accuracy and 0.83 macro F1 on the provided 959-image test set. Its performance should not be compared directly with clinical dermatology systems without accounting for dataset provenance, image capture, target labels, and clinical reference standards. The training pipeline and evaluated SVM differ from the currently supplied standalone Keras serving code.

Compared with ML triage in emergency departments [18], [19], the risk model uses no vital signs or laboratory values, which makes it usable in shelters without equipment, but it also limits its maximum accuracy. Compared with existing systems (Table 2.2), the module is the only one that provides individual-level, exposure-aware risk triage and skin-image assessment inside a flood-response platform.

7.3 Research contribution

The research contributions of this thesis are: (1) a flood-specific feature schema that combines symptoms, symptom duration, flood exposures and risk factors for individual risk triage, together with empirical evidence of its predictive value; (2) an

evaluation of transfer-learning CNNs for classifying common post-flood skin rashes from ordinary photographs; and (3) a conceptual framework (Figure 2.1) that links individual health-risk assessment to disaster-response operations, addressing the data, modality and integration gaps identified in Section 2.5.

7.4 Product contribution

The product contribution is a pair of deployable, documented inference services: a FastAPI rash-classification microservice with health checking, input validation and structured probability output, and a schema-guarded, version-checked risk prediction bridge. Both are integrated into the R26-IT-031 platform. Their design choices, including preprocessing stored inside the model, fail-fast loading and explicit error codes, make the services safe to replace, scale and maintain.

7.5 Limitations and threats to validity

Testing revealed challenges in distinguishing health conditions with overlapping symptoms or similar visual characteristics. The initial unit test for the disease-management module failed, and the associated defect was classified as high severity. Although subsequent regression testing reported successful differentiation of the tested cases, the available documentation does not provide sufficient case-level evidence to establish the clinical reliability of the classification results.

Furthermore, system testing was conducted under controlled conditions rather than in a real-world flood emergency. Consequently, the findings demonstrate the functional feasibility of the proposed system but do not establish its effectiveness in operational disaster-response settings. Further validation using representative clinical datasets, assessments by qualified healthcare professionals, and reproducible end-to-end integration testing is necessary before the system can be considered for deployment in operational health-screening environments.

7.6 Implications

For practice, the module shows that a low-cost web tool can help medical teams prioritise people after a flood using information that field officers can collect without

equipment, and that health information can be connected to rescue and relief operations, as recommended by the Sendai Framework's emphasis on health resilience [44]. For research, the results support collecting locally labelled, outcome-linked post-flood health data in Sri Lanka, which would allow stronger models and proper clinical validation. For policy, any deployment must be governed by the PDPA and the WHO guidance on AI for health [7], [34], with clear human accountability for clinical decisions.

8 COMMERCIALISATION AND SUSTAINABILITY

8.1 Value proposition and target users

The module offers early identification of flood-related health risks, prioritisation of scarce medical attention, and an area-level view of emerging health problems, all inside the same platform used for flood warning, rescue and relief. The target users are: (1) government agencies such as the Disaster Management Centre, the Ministry of Health, Regional Directors of Health Services and Medical Officers of Health; (2) local authorities that run safety centres; (3) humanitarian NGOs and international agencies such as the Red Cross and UN agencies; and (4) field officers and volunteers who collect data in affected communities.

8.2 Market and competitive position

As shown in Table 2.2, existing disaster platforms focus on coordination and logistics, and disease-surveillance tools work at population level. The module's competitive advantages are exposure-aware individual triage, image-based rash assessment and integration with the rest of the flood-response workflow. The main competitive risks are general-purpose health information systems already used by the Ministry of Health, such as DHIS2-based platforms, and consumer dermatology applications. Interoperability, for example exporting summaries to existing health information systems, is therefore more realistic than replacing them.

8.3 Business or sustainability model

Three models are proposed: (1) a **government or institutional licence** for national and provincial agencies; (2) a **software-as-a-service (SaaS) subscription** for NGOs and local authorities, priced by number of users or deployments; and (3) **donor-sponsored deployment**, in which humanitarian organisations fund the system for disaster-prone areas, with discounted pricing for low-income settings. Table 8.1 shows the estimated development budget from the project proposal.

Table 8.1: Estimated development budget

Category	Estimated cost (LKR)
Hardware (development machine use, smartphone camera, storage)	31,130
Software (open-source tools)	0

Category	Estimated cost (LKR)
Cloud services (hosting, storage)	62,261
Human effort	60,000
Miscellaneous	31,130
Total	183,521

Sustainability depends on keeping running costs low (open-source software and stateless, containerised services that scale up only during flood events), on retraining the models regularly with locally collected data, and on maintenance by a partner institution after the project.

8.4 Scalability, intellectual property and compliance

Scalability. Both services are stateless and can be scaled horizontally behind a load balancer when demand peaks during a flood. **Intellectual property.** The source code, trained models and the curated feature schema are the main intellectual property and can be protected by copyright and licence agreements; third-party datasets and pre-trained weights must be used according to their licences. **Compliance.** Deployment requires compliance with the PDPA [7], including lawful processing, purpose limitation and data-subject rights, and with Ministry of Health requirements. Because the tool supports clinical decisions, it is positioned as a triage aid that requires professional confirmation, and wider clinical use would need appropriate regulatory review.

8.5 Contribution to the Sustainable Development Goals

The component contributes to the United Nations Sustainable Development Goals [45]: **SDG 3 (Good Health and Well-being)**, through earlier detection and treatment of post-flood disease (target 3.d on early warning and management of health risks); **SDG 11 (Sustainable Cities and Communities)**, by reducing the human impact of water-related disasters (target 11.5); and **SDG 13 (Climate Action)**, by strengthening resilience and adaptive capacity to climate-related hazards (target 13.1).

9 CONCLUSION AND FUTURE DIRECTIONS

9.1 Conclusion

This thesis investigates an integrated post-flood health-risk decision-support module with a symptom-based Low/Medium/High risk classifier and an image-classification service whose supplied JSON configuration lists eight labels: Cellulitis, Dengue, Dermatitis, Eczema, Impetigo, Leptospirosis, Ringworm, and Scabies. The inference interfaces are described, but eight-output model compatibility, held-out predictive performance, service latency, and stakeholder validation need authentic supporting evidence. The outputs are screening aids, not medical diagnoses.

9.2 Achievement of objectives

Table 9.1: Achievement of objectives

Objective	Evidence	Status
O1 Identify relevant conditions, symptoms, exposures and risk factors	Chapter 2; Section 3.2; Appendix A	Partly evidenced: input feature schema
O2 Develop and evaluate the risk model	Sections 3.3 and 6.1.1	Model inference evidenced; evaluation unverified
O3 Develop and evaluate the rash CNN	Sections 3.3 and 6.1.2	Model inference evidenced; evaluation unverified
O4 Deploy reliable inference services	Sections 4.4, 5.2 and 5.4	Implemented code supplied; tests unverified
O5 Integrate alerts and dashboard	Sections 5.3 and 6.4	Integration planned; final results unverified
O6 Evaluate performance and usefulness	Chapter 6	Not verified

9.3 Individual research and product contributions

Individually, the author contributed the problem analysis and feature schema for post-flood triage, the training and evaluation of both models, the FastAPI rash-classification

microservice, the schema-guarded risk prediction bridge, the integration contract with the shared backend, and the testing and stakeholder validation of the component (Section 1.8).

9.4 Future directions

- Collect locally labelled, consented post-flood data in Sri Lanka, including clinical outcomes, to retrain and clinically validate both models.
- Add a clinically reviewed "uncertain / refer" outcome, explicit safeguards for non-specific disease labels (including dengue and leptospirosis), out-of-distribution checks and probability calibration; extend the model only after collecting defensible reference labels.
- Evaluate and reduce performance differences across skin tones [15], [16].
- Provide explanations for clinicians using SHAP [42] for the risk model and Grad-CAM [43] for the CNN.
- Support Sinhala and Tamil, offline data capture with later synchronisation, and on-device inference (e.g. TensorFlow Lite) for areas with poor connectivity.
- Harden the services for production: restricted CORS, upload limits, rate limiting, lifespan start-up handlers, a persistent risk-model service, containerisation and monitoring.
- Combine individual risks with flood-extent data from Component 1 to detect spatial clusters and give early outbreak warnings.

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GLOSSARY

Confidence score: The softmax probability that the rash classifier assigns to its predicted class; it is not a calibrated clinical probability unless calibration is applied.

Dermatitis: Inflammation of the skin, including irritant and contact dermatitis caused by prolonged contact with floodwater.

Eczema: A chronic or relapsing inflammatory skin condition that can flare after exposure to irritants, moisture and heat.

Flood exposure: Any reported contact with flood conditions, such as wading through floodwater, drinking unsafe water or staying in a crowded shelter.

Leptospirosis: A bacterial zoonosis spread through water or soil contaminated by the urine of infected animals; it is a major post-flood disease in Sri Lanka.

Macro F1-score: The unweighted mean of per-class F1-scores, which treats each class equally regardless of its size.

Microservice: A small, independently deployable service that exposes one capability through a network API.

Ringworm (tinea): A superficial fungal skin infection that spreads easily in warm, humid and crowded conditions.

Transfer learning: Re-using a network trained on a large dataset (such as ImageNet) as the starting point for a related task with less data.

Triage: The process of ranking people by urgency so that limited medical resources reach the most urgent cases first.

APPENDICES

Appendix A Research instruments and extended results

A.1 Input schema (confirmed): age; gender; symptom_duration; selected_symptoms; flood_exposures; risk_factors. Multi-select fields are serialised using semicolon-and-space separators. Full option lists require options.json.

A.2 Participant-facing research documents: no signed consent form, information sheet or stakeholder questionnaire was supplied. Add the approved originals only if they exist.

A.3 Suggested reproducible API checks: verify the health endpoint, valid and invalid uploads, symptom readiness probe, three-level symptom risk output, eight-item image label configuration, Keras output shape matching eight classes, and correct class-index mapping. Attach execution logs and actual pass/fail records.

A.4 Extended experimental results: not provided; append the actual fold scores, confusion matrices, training curves and misclassified examples exported from the training notebooks.

A.5 Model card summary [35]

Table A.1: Model card summary

Item	Risk model	Rash CNN
Intended use	Triage support for post-flood health screening	Triage support for common post-flood rashes
Not intended for	Diagnosis; use without professional review	Diagnosis; images other than skin rashes
Inputs	Age, gender, duration, symptoms, exposures, risk factors	RGB image, resized to 224 × 224
Outputs	Low / Medium / High	Eight configured labels (see class_names.json); scores require verification
Training data	Dataset source and size not provided	Dataset source and size not provided
Known limitations	Labels not outcome-confirmed	Closed-set; possible skin-tone bias; uncalibrated confidence